AI for angiographic sequence analysis

Systematic review of 120 studies (PRISMA 2020)

Study selection

5,064 records screened

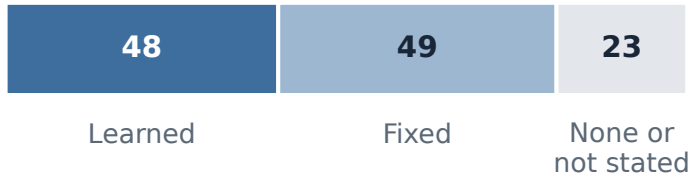
122 eligible

120 included

LLM screening, validated by two blinded human reviewers (Cohen's $\kappa = 0.98$)

How studies model time

80.8% link information across frames



Of these, half rely on fixed operations such as stacking, registration or curve fitting. Only 8.3% use Transformer or state-space models, all since 2023.

12 research gaps in 4 clusters

Data infrastructure

- G1 No public benchmarks
- G2 Generalizability
- G5 Annotation scarcity
- G10 Privacy-preserving

Modeling & learning

- G3 Temporal modeling
- G4 2D constraints
- G9 Class imbalance

Clinical deployment

- G6 Clinical validation
- G7 Explainability
- G8 Multi-modal integration

Outcome & reporting

- G11 Outcome prediction
- G12 LLMs & reporting